

TOPS TECH Task 3 (CS)

Task-3 Cyber Security threats

1. **List 5 different types of cyber security threats you might encounter while using popular apps like Instagram, WhatsApp, or Zomato, and give a real-life example for each.**

Ans:

Different Types of Cyber Security Threats

1. Phishing

- Phishing is a common cyberattack where attackers pretend to be a trusted person or company to trick users into sharing sensitive information such as passwords, OTPs, or banking details.

Real-World Example:

- A user receives a message claiming to be from Instagram saying that their account will be suspended.
- The message contains a login link. When the user enters their credentials, the attacker steals the account information.

2. Malware

- Malware refers to harmful software that can enter a device without the user's knowledge and perform unwanted actions such as stealing data, damaging files, or monitoring activities.

Real-World Example:

- A user downloads a free application from an untrusted website.
- After installation, the malware starts collecting personal information from the device.

3. Ransomware

- Ransomware is a dangerous cyber threat that locks or encrypts important files and prevents users from accessing them until a ransom is paid.

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Real-World Example:

- A company employee opens a malicious email attachment.
- The ransomware encrypts all company files and demands money to restore access.

4. Distributed Denial of Service (DDoS) Attack

- In a DDoS attack, attackers use multiple systems to send a massive amount of traffic to a website or application, making it slow or completely unavailable.

Real-World Example:

- An online food delivery platform such as Zomato experiences unusually high malicious traffic, causing users to face difficulties accessing the service.

5. Identity Theft / Account Hacking

- Identity theft occurs when cybercriminals gain unauthorized access to someone's personal information or online accounts and misuse them.

Real-World Example:

- A hacker gains access to a user's WhatsApp account and sends fake messages to friends and family asking for money.
- Cyber security threats can affect anyone who uses the internet. Understanding these threats and following basic security practices such as using strong passwords, enabling two-factor authentication (2FA), and avoiding suspicious links can help users stay safe online.

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2. Simulate a phishing attack scenario by designing a fake login page for an app you use (for example, a mock Instagram or Flipkart login screen in HTML). Explain how a user could be tricked and what signs would help spot the threat.
Hint: Do not collect or submit any real credentials—just create the page design and description.

Ans:

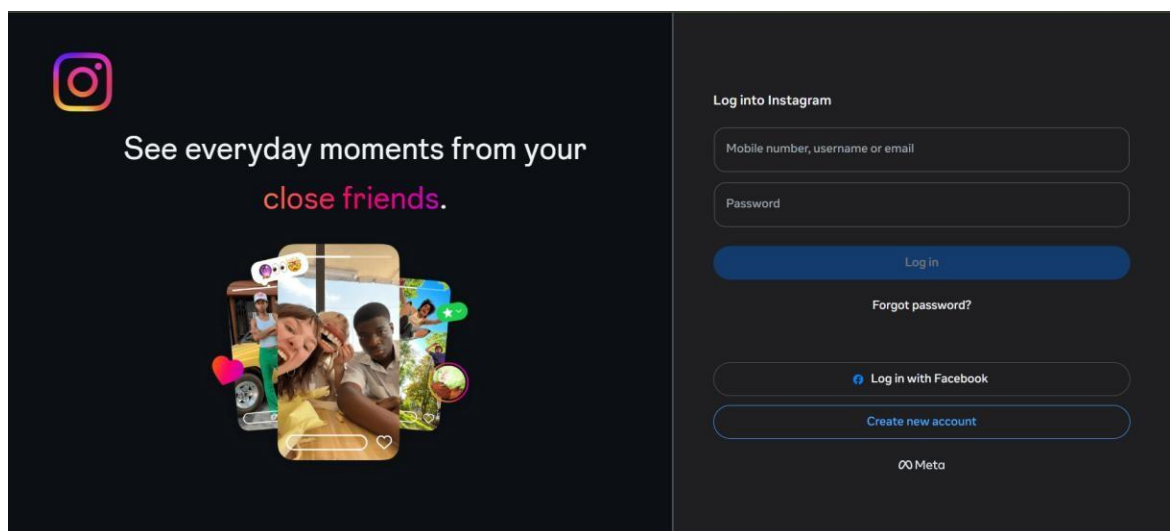
Fake Login Page Simulation (Instagram)

Objective

- The purpose of this task is to understand how phishing attacks work by creating a fake login page for educational purposes.
- This activity helps users identify fake websites and learn how attackers try to steal login credentials.

Fake Login Page Chosen

Instagram Login Page



This fake login page is designed to look similar to the original Instagram login page. It contains fields for username, password, and a login button.

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How a User Could Be Tricked

- A cybercriminal may send a link through email, SMS, or social media claiming it is the official Instagram login page.
- Since the page looks real, users may enter their login credentials. The attacker can then steal this information and misuse the account.

Warning Signs of a Fake Login Page

- The website URL is different from the official website
- Spelling or grammar mistakes on the page
- No HTTPS security lock
- Login link received from unknown sources
- Slightly different design from original website
- Asking for unnecessary information

Prevention Methods

- Always verify the website URL before logging in
- Do not click on unknown links
- Enable Two-Factor Authentication (2FA)
- Use strong and unique passwords
- Keep browser and device updated

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3. **Research and write a short report (150-200 words) on a recent cyberattack that affected a major Indian tech platform (such as Paytm, IRCTC, or PhonePe). Describe the type of threat, how it was carried out, and what was done to fix it.**

Ans:

Cyberattack Case: IRCTC Security Incident

1. Platform Affected

- IRCTC (Indian Railway Catering and Tourism Corporation)

2. Type of Threat

- Credential stuffing attack
- Account hacking
- Unauthorized login attempts

3. How the Attack Was Carried Out

- Attackers used leaked usernames and passwords from other websites
- Automated tools were used to try multiple login combinations
- Users who reused passwords were targeted
- Successful logins gave attackers access to accounts

4. Impact of the Attack

- Unauthorized access to user accounts
- Misuse of accounts for ticket booking
- Risk of personal data exposure

5. Fix / Actions Taken

- Strong password policies introduced
- Suspicious login monitoring increased
- Two-factor authentication (2FA) encouraged
- Blocking of suspicious IP addresses
- Improved security monitoring systems

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4. **Pick any one cyber security threat (like ransomware, phishing, or DDoS) and create a simple infographic or poster (hand-drawn or digital) that explains how it works, who is at risk, and how to prevent it.**

Ans:

Cyber Security Infographic / Poster Content:

Phishing Attack

1. What is Phishing?

- A cyberattack where fake messages, emails, or websites are used
- Trick users into sharing passwords, OTPs, or personal data
- Looks like real companies (Instagram, Bank, Google, etc.)

2. How it Works

- Attacker creates a fake login page or message
- Sends link via email, SMS, WhatsApp, or social media
- User clicks the link and enters details
- Attacker steals login credentials
- Account gets hacked or misused

3. Who is at Risk?

- Social media users (Instagram, Facebook, WhatsApp)
- Online banking users
- Students and general internet users
- Anyone who clicks unknown links

4. Warning Signs

- Unknown or suspicious links
- Spelling mistakes in message or website
- Urgent messages like “account will be blocked”
- Fake website URL
- Asking for OTP or password

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5. How to Prevent It

- Do not click unknown links
- Always check website URL carefully
- Enable Two-Factor Authentication (2FA)
- Do not share OTP or password
- Use strong passwords
- Verify sender before trusting messages



PHISHING ATTACK

Don't get hooked. Stay alert, stay safe!

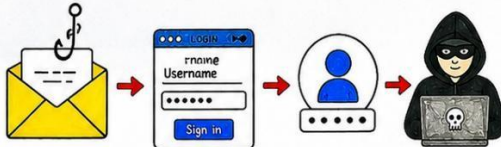


WHAT IS PHISHING?

Phishing is a type of cyber attack where attackers pretend to be a trusted person or organization to trick you into sharing sensitive information like usernames, passwords, OTPs or bank details.



HOW IT WORKS



1. Attacker sends fake email, SMS or message.
2. You click on the link and land on a fake website.
3. You enter your login details or other information.
4. Attacker steals your information and misuses it.

WHO IS AT RISK?

- Social media users
- Online banking users
- Students
- Employees
- Anyone who uses the internet



WARNING SIGNS

	Unknown or suspicious links
	Spelling and grammar mistakes
	Urgent messages like "Your account will be blocked" or "Verify now"
	Website URL looks different
	Asking for OTP, password or personal information

HOW TO PREVENT?

- ✓ Do not click on unknown links.
- ✓ Always check the website URL carefully.
- ✓ Enable Two-Factor Authentication (2FA).
- ✓ Do not share OTP, password or personal information.
- ✓ Use strong and unique passwords.
- ✓ Verify the sender before trusting any message or email.



BE AWARE. BE ALERT. BE SAFE. 